



Optical coherence tomography in the assessment of oral squamous cell carcinoma resection margins



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ABSTRACT

Background: Incomplete surgical removal of cancer is believed to be the main cause of local recurrence and high mortality. This study assessed the use of optical technology (namely optical coherence tomography [OCT]) in examining oral squamous cell carcinoma (OSCC) resection margins to assess if this modality could guide the surgeon during surgical resections.

Materials and methods: Twenty-eight T₁–T₂ N0M0 oral squamous cell carcinoma patients participated in this study. Nineteen patients were males and nine were females. The majority of lesions were in the ventro-lateral tongue, floor of mouth, retromolar trigone and the buccal mucosa.

Following tumour resection, the specimen resection margins were optically scanned in the immediate ex vivo phase. Two independent assessors commented on the four resection margins of each specimen. The findings were then compared to the corresponding gold standard histopathology. The average epithelial thickness for both tumor-free and tumor-involved margins was also calculated.

Results: The pathology reports of the 112 margins revealed 90 tumor-free margins and 22 tumor-involved margins. Examining the data from both senior operating surgeons (assessors), the overall sensitivity and specificity was found to be 81.5% and 87%, respectively. Whilst the positive predictive value was 61.5% and the negative predictive value was 95%. OCT accuracy for the first assessor was 88% and for the second assessor 84%. The assessors' inter-observer agreement was "very good" for superior, inferior and medial margins; while agreement on the lateral surgical margin status was "good". Using OCT, the mean epithelial thickness at the tumor-free resection margins was 360 μm; while, it was 567 μm for the tumour-involved margins.

Conclusion: OCT is a valuable tool in the assessment of surgical margins. Tumour-involved margins can be identified by architectural changes and increase in epithelial layer thickness on the OCT image. Further studies are required to assess tumour margins in vivo.

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1. Introduction

The UK guidelines judge both mucosal and deep margins of ≥ 5 mm free of tumour as clear, 1–5 mm as close and ≤ 1 mm as involved. This usually ignores the formalin-shrinkage effect, which can reach up to 30%. So in order to achieve a 5 mm pathological clearance, 8–10 mm in situ surgical margin, a considerable amount of normal tissue needs to be taken [1].

Positive (tumour-involved) or close margins are associated with increase in local recurrence with an attendant negative effect on survival. Although there is lack of high evidence-based publications that show a significant correlation between close resection margins and negative outcomes when it comes to survival and recurrence, several studies have shown that local recurrence and overall survival benefit from achieving negative resection margins [1–4].

One of the unresolved clinical issues is the absent of any objectively reliable and accurate assessment modality for resection margin status. In most cases, intraoperative visual inspection and palpation is the standard approach. Unfortunately, this technique resulted in positive or close surgical margins in almost 40% of the patients, in one study [2]. Intraoperative frozen sections, although widely used, has its own inherited problems. This technique is

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costly, time consuming. Additionally, the frozen samples do not represent the whole margins but random sampling; which may misrepresent the real situation [3,4]. To help overcome these clinical challenges, a highly sensitive, non-invasive, cost-effective and in vivo diagnostic tool is required.

The use of optical technology (diagnostics) in clinical trials has increased in the past few years. If found to be effective; these tools can help by reducing pathology workload and by acting as adjunct to pathology in specific situations. Optical coherence tomography (OCT) is one of these diagnostic tools that hold great promise. Huang et al. have pioneered the principle of OCT in 1990s [5]. Its clinical use was, initially, restricted to the field of ophthalmology [6]. Soon after, its clinical application expanded to involve several body tissue pathologies [7–9]. In head and neck, OCT has been used to detect early epithelial changes in the oropharyngeal/laryngeal region. Prestin et al. [10] in a study measuring oral tissue epithelium using OCT found that it was feasible to identify oral malignancy and determine the approximate grade of dysplasia.

In a preliminary immediate ex vivo study carried out by our group on oral tissues, we compared findings of optical coherence tomography (OCT) with histopathological results of suspicious oral lesions. Thirty-four oral lesions from 27 patients were subjected to swept-source frequency-domain OCT. OCT could identify diseased areas but could not provide a diagnosis or differentiate between abnormal lesions of close pathological characteristics (i.e. different grades of dysplasia and carcinoma in situ). Overall, the pilot study, by Jerjes et al., confirmed the feasibility of using OCT to identify architectural changes in malignant tissues [11].

Previous studies using different OCT systems confirmed the effectiveness of such diagnostic technology but the generated images were of low resolution due to poor penetration depth and lateral resolution. Wilder-Smith et al. [12] evaluated the clinical capability of non-invasive in vivo OCT for diagnosing oral dysplasia and malignancy in 50 patients. Their data demonstrated good capability of the in vivo OCT but have suffered from resolution problems. An earlier study by Ridgway et al. [13] outlined future applications of OCT technology by examining the mucosa of the oral cavity and oropharynx in 41 patients during operative endoscopy. Optical coherence tomographic imaging showed distinct zones of normal, altered, and ablated tissue microstructures for each pathologic process studied.

In laryngology, OCT was found to be efficient, quick, and reliable imaging modality in guiding surgical biopsies, intraoperative decision making, and therapeutic options of various laryngeal pathologies [14]. A prospective study by Just et al. [15] assessed OCT application in benign and dysplastic laryngeal epithelial lesions. They showed that the thickness of the epithelium is the main criterion for degree of dysplasia. In their study, OCT provided test outcomes for differentiation between benign laryngeal lesions and dysplasia/CIS with sensitivity of 88% and specificity of 89%. However, and due to the limited penetration depth of the laser light in hyperkeratotic lesions, the basal cell layer was no longer visible, preventing reliable assessment of such lesions.

An interesting study by Kraft et al. [16] using microlaryngoscopy with OCT during the diagnostic investigation and intraoperative monitoring of laryngeal disease presented a higher sensitivity in predicting invasive tumor growth and epithelial dysplasia, when compared to microlaryngoscopy alone. A recent pilot study by Conti de Freitas et al. [17] looked at resection margins and demonstrated that OCT is capable of recognizing and differentiating neck tissues encountered during thyroid and parathyroid surgeries.

The aim of this immediate ex vivo clinical study was to assess the role of OCT in identifying tumour-involved resection margins in patients undergoing surgery for oral squamous cell carcinoma (OSCC). Epithelial layer thickness and architectural changes were the main parameters assessed at each of the resection margins.

Table 1

Demographics of the cohort included in this study.

	No. (%)		No. (%)
Gender		Symptoms	
Male	19 (67.9)	Asymptomatic	19 (67.9)
Female	9 (32.1)	Pain	4 (14.3)
		Bleeding	5 (17.9)
Location		Smoking status	
Ventro-lateral tongue	7 (25.0)	Current smoking	10 (35.7)
Floor of mouth	6 (21.4)	Ex-smoker	12 (42.9)
Retromolar trigone	4 (14.2)	Non-smoker	6 (21.4)
Buccal mucosa	3 (10.7)	Drinking status	
Lower lip	2 (7.1)	Current drinker	12 (42.9)
Hard palate	2 (7.1)	Ex-drinker	6 (21.4)
Upper lip	2 (7.1)	Non-drinker	10 (35.7)
Soft palate	2 (7.1)		
Colour		Pan chewing	3 (10.7)
Leukoplakia	4 (14.3)	Diagnosis	
Erythroplakia	14 (50.0)	T ₁ disease	20 (71.4)
Speckled leukoplakia	10 (35.7)	T ₂ disease	8 (28.6)
Clinical features		Resection	
Plaque	6 (21.4)	CO ₂ Laser	7 (25.0)
Papule	6 (21.4)	Surgical resection	17 (60.7)
Ulcer	16 (57.1)	Electrosurgical	4 (14.3)
Medical history			
ASA I	20 (71.4)		
ASA II	8 (28.6)		

2. Materials and methods

Identical protocols were used to recruit 28 consecutive patients who presented with suspicious oral lesions to the UCLH Head and Neck Centre, London between 2008 and 2010. The study protocol was approved by the Committee of the Ethics for Human Research.

Nineteen patients were males (67.9%) and nine were females (32.1%); with a mean age of 61 years (range 36–103 years). Approximately one-third of the patients were current smokers. Over 40% of the patients consumed alcohol on a regular basis and less than 10% chewed betel nut (Table 1). Half of the lesions were ulcers; and the rest manifested as plaques or papules. Fifty percent ($n=14$) of the lesions manifested as erythroplakia, 35.7% presented as leukoerythroplakia and 14.3% as homogeneous leukoplakia. The anatomical distribution of the lesions showed 7 in the ventro-lateral tongue, 6 in the floor of mouth, 4 in retromolar trigone and 3 in the buccal mucosa (Table 1).

An incisional surgical biopsy was acquired from each patient and confirmed the diagnoses of oral squamous cell carcinoma (OSCC). Clinical staging at time of presentation showed that 20 patients had T1N0 disease [mean depth of invasion 4.6 mm, minimum 1.2 mm, maximum 6.3] and 8 patients had T2N0 disease [mean depth of invasion 5.8 mm, minimum 1.4 mm, maximum 9.6]. All patients were discussed at our multi-disciplinary meeting, and the ones with deep malignancies invading >5 mm, although with clinical N0 necks, were recommended for further intervention in the form of selective neck dissection to ensure loco-regional control. The presence of moderate or severe dysplasia at margin is strongly correlated with inferior local control and potentially worse disease-free survival and hence [18], any patient in our study that fits this description received postoperative radiotherapy and/or chemotherapy as a result.

A “pre-made proforma” was used to collect clinic-pathological and optical data from each patient included in this study. We

use this electronic form in our head and neck unit mainly for audit and research purposes. It includes several sections: general demographics, clinical, pathological, radiology and research data. The proforma has been designed according to the clinical protocols at University College London and defined to maintain patient confidentiality. The researchers were very accurate in record keeping which later on facilitated statistical analysis and study results.

All surgical resections were performed under general anaesthesia. The resections were preserved in normal saline and within 1 hour (immediate ex vivo) were subjected to optical coherence tomography scanning. All specimens were then processed for assessment of the surgical margins status. Histopathological assessment was carried out by one oral and maxillofacial pathologist to ensure objectivity. Assessment was conducted according to the World Health Organisation guidelines. The pathologist assessed all the surgical margins of all specimens (four per specimen). No further resection margins were sent to the pathologist.

The process of co-localization was described in a previous descriptive study published by our group [11]. The co-registration process involved diagrams, digital images and specimen orientation using sutures and special ink. For each surgical specimen, OCT images were acquired from the four resection margins (superior, inferior, medial and lateral), (Fig. 1). Each margin was scanned as a whole, however the “specimen holding plate” was not designed to allow the continuous scanning of all four resection margins in one go; any attempt to do this manually resulted in image distortion.

Assessment of OCT images was carried out by two surgeons; both were taught about interpreting OCT images and provided with training set to consolidate their knowledge and assessment skills prior to the assessment of the 112 OCT images (Fig. 2). Each assessor was provided with brief clinical history of the patient and was asked to comment on every corresponding OCT image using the pre-made proforma. Mean values of a number of metric readings were recorded for the epithelial layers at the resection margins area.

In general, an abnormal lesion is identified on OCT when there is loss of well-defined tissue layers. The technology is fairly new and new algorithms are being developed to improve the reading of these optical signatures, including the attempt to analyse the changes in the OCT signal to determine blood perfusion and tissue organisation. The criterion that was used to diagnose cancer was based on identifying architectural changes and any invasion of the basement membrane. OCT images usually provide information about tissue layers thickness, which was also helpful in the assessment. Four resection margins were assessed per patient; the assessors were asked to comment on the architectural changes in tissue and the status of the basement membrane (intact or breached). Identification of the increase in cellular aggregation and disorganisation, irregular collagen or elastin fibres arrangements, micro-anatomical structures invasion (i.e. vessels, tongue papillae, rete pegs, ..., etc.) and the disorganisation of the basement membrane are usually indicative of cancerous process. Increase in keratin and epithelial layers thickness usually supports the diagnosis, as well as lamina propria disorganisation. The kappa scoring system was used to assess the agreement between the two readers on all the four surgical margins as well as the status or integrity of the basement membrane.

We used a swept-source frequency-domain optical coherence tomography microscope (Michelson Diagnostics EX1301 OCT Microscope V1.0), the components of which are illustrated in previous study by our group [11]. The light source used is a Santec HSL-2000, with an imaging wavelength of 1310 nm, axial optical resolution of <10 μm , and lateral optical resolution of <10 μm . The

Table 2

Histopathological status of the four resection margins of the cohort.

Superior margin	No. (%)
Free	24 (85.7)
Involved	4 (14.3)
Inferior margin	
Free	22 (78.6)
Involved	6 (21.4)
Medial margin	
Free	21 (75.0)
Involved	7 (25.0)
Lateral margin	
Free	23 (82.1)
Involved	5 (17.9)

Table 3

Measurement of resection margins using OCT. KL: Keratin layer, EL: epithelial layer.

	Range <i>b</i> (μm)	Minimum (μm)	Maximum (μm)	Mean (μm)
Tumour free margins				
KL thickness	17	15	32	21
EL thickness	290	190	480	390
Tumour involved margins				
KL thickness	26	11	37	24
EL thickness	238	530	760	567

system provides an image resolution of 5.3 μm /pixel with a maximum image width of 6 mm, a sub-surface imaging depth of 1.5 mm, and a focal depth of 1 mm. Samples can be manipulated to see the full quality results on the screen instantly, with an image capture time of <100 ms and refresh rate of >1 Hz. Images were acquired in two modes: black and white, and inverted colour mode. We used the black and white mode in OCT to assess the images and identify pathological processes. We believe that, from previous experience, there is no difference between this mode and inverted colour mode.

3. Statistical analysis

The overall sensitivity and specificity for interrogating the 112 resection margins were calculated using SPSS package (Statistical Package for the Social Sciences, Chicago, IL). The Inter-observer differences using Kappa and positive and negative predictive values were measured using the same software. Mean values of the epithelial thickness of the resection margins were obtained. Then comparison was made between positive and negative margins. Chi-square testing was used to confirm this relationship and identification of the *P*-value.

4. Results

The pathological results of the margins status revealed 90 (80.35%) defined as tumor-free and 22 (19.65%) as having some degree of tumor (Table 2). The resection margin containing tumour may be partially or completely involved. On OCT, tumour-involved margins displayed sudden change in epithelial thickness or partial loss of basement membrane integrity (Figs. 3–5). The mean thickness of the epithelial layer of the tumour-free margin was 360 μm , and for the tumour-involved margin 567 μm (Table 3). The relationship was significant ($P < 0.001$).

The sensitivity and specificity for the first reader was 86% and 89%, respectively. Whilst the positive predictive value (PPV) was 66% and the negative predictive value (NPP) was 96%. The accuracy of OCT was 88% (Table 4). There were 7 false positives and 3 false negatives, with a positive likelihood ratio (LR+) of 7.16 and negative likelihood ratio (LR-) of 1.13. The sensitivity and specificity for the

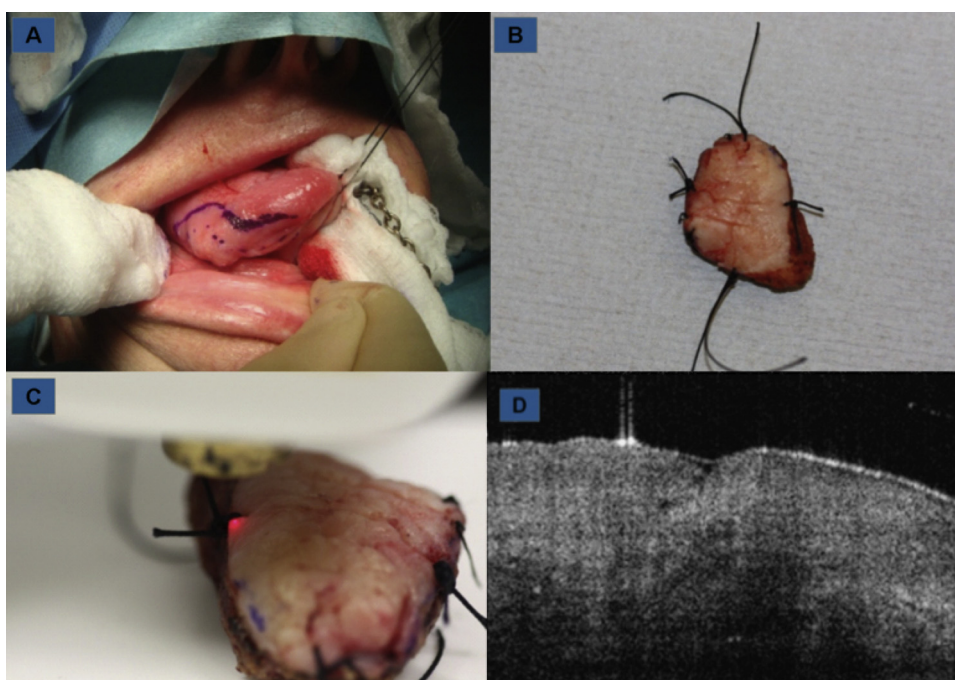


Fig. 1. Resection margins assessment using OCT. A: Marking tumour prior to resection. B: Tumour resected and specimen oriented with sutures and special dyes. C: Resected specimen subjected to OCT. D: OCT scanning and image generation.

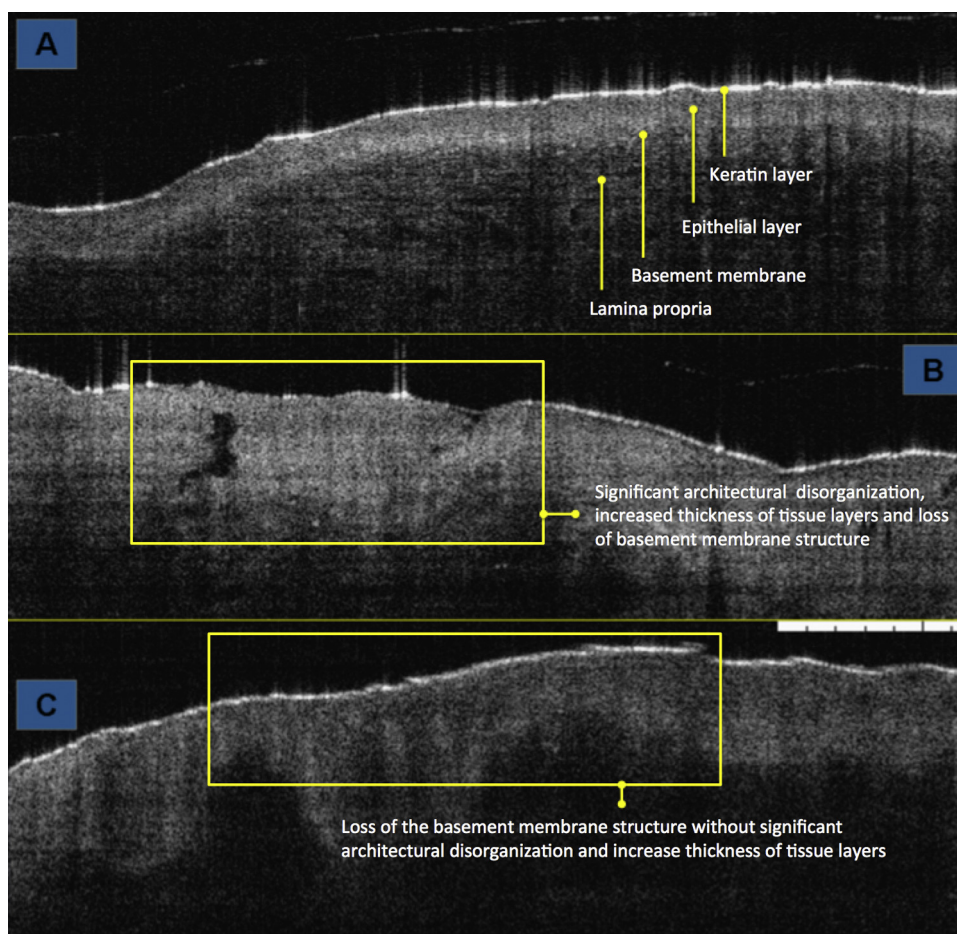


Fig. 2. Data (OCT images) used in training both clinicians in identifying pathological processes. The images were compared to the gold standard histopathology. A: Tumour free margin; B: tumour 1–5 mm from the margin; C: tumour ≤ 1 mm from the margin.

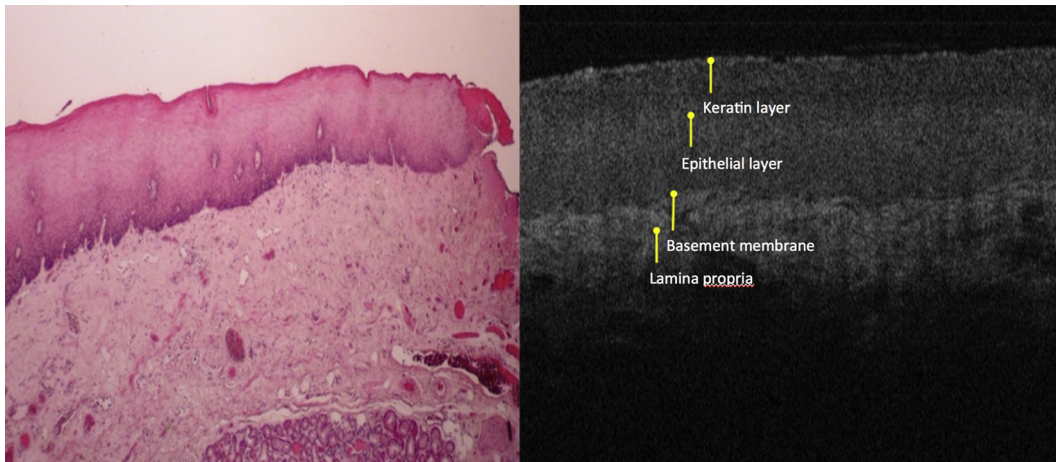


Fig. 3. Buccal mucosa resection margin showing no tumour elements and intact basement membrane.

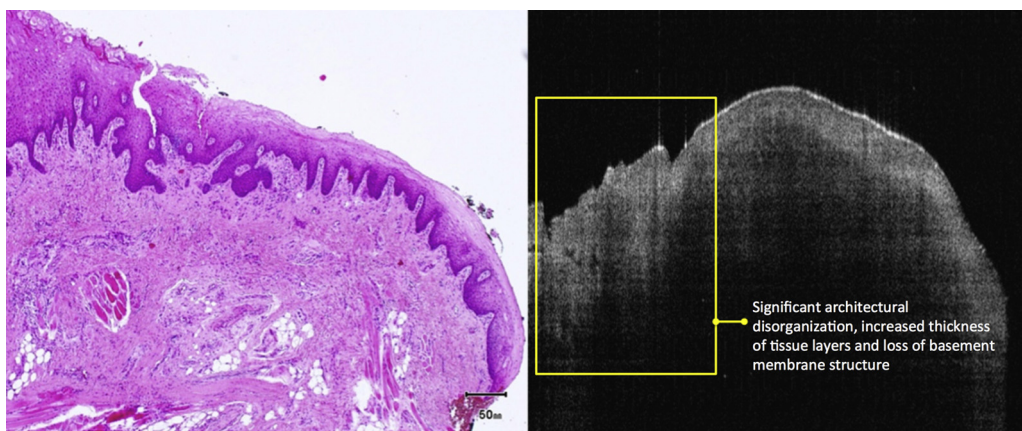


Fig. 4. Floor of mouth mucosa resection margin showing partial tumour involvement (within the 5 mm edge) and breached basement membrane.

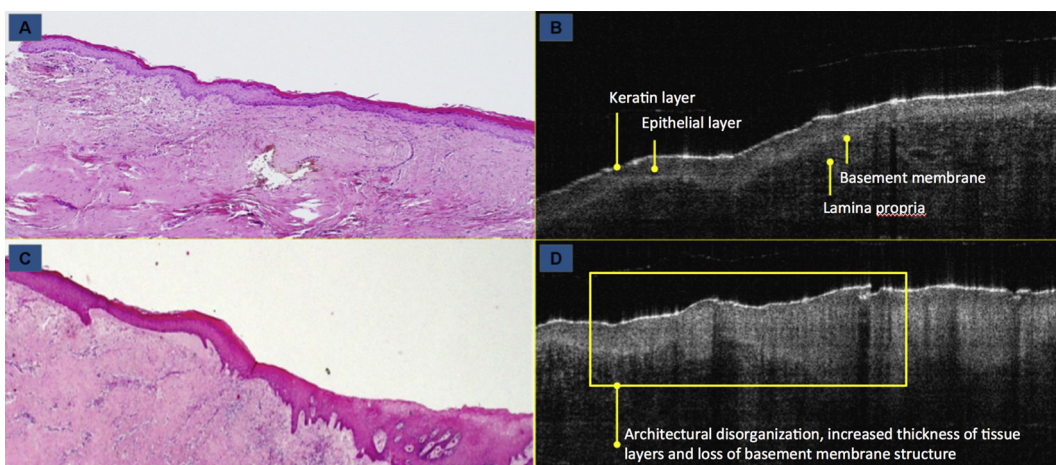


Fig. 5. A + B: tumour free margin with intact basement membrane from oral cancer lesion; C + D: cancer involved margin with breached basement membrane.

second reader was 77% and 86%, respectively. The PPV was 57% and the NPP was 94%. The accuracy of OCT was 84% (Table 5). There were 13 false positives and 5 false negatives, with a LR+ of 5.5 and LR- of 1.16.

The average sensitivity and specificity was 81.5% and 87%, respectively. The PPV and NPP was 61.5 and 95, respectively. The accuracy was 86%. The inter-observer agreement was “very good” when assessing all margins except the lateral margin (0.644), (Table 6).

5. Discussion

The precise assessment of tumour in its vertical and horizontal extensions can be problematic and this is paramount to ensure complete removal and to predict local or regional failures [19,20]. The potential of OCT in objective disease assessment of tumour margins has been explored. Although the tumour-involved margins sample was small, our results indicated that OCT could be used to examine

Table 4

Clinical assessment by the first reader. Sensitivity 86%; specificity 88; positive predictive value (PPV) 66%; negative predictive value (NPV) 96%; accuracy of OCT 88%. TP: True positive, FP: false positive, TN: true negative, FN: false negative.

		Histology		
		Positive	Negative	Total
OCT	Positive	19	10	29
	Negative	3	80	83
	Total	22	90	112

Table 5

Clinical assessment by the second reader. Sensitivity 77%; specificity 86; positive predictive value (PPV) 57%; negative predictive value (NPV) 94%; accuracy of OCT 84%. TP: True positive, FP: false positive, TN: true negative, FN: false negative.

		Histology		
		Positive	Negative	Total
OCT	Positive	17	13	30
	Negative	5	77	82
	Total	22	90	112

Table 6

Raw kappa scores showing levels of un-weighted agreement between the two readers.

	Kappa
Superior margin	0.806
Inferior margin	0.713
Lateral margin	0.644
Medial margin	0.718

surgical margins in the oral cavity and identify the involved ones.

The inclusion criteria included only T₁–T₂ OSCC with no nodal disease, which was aimed to facilitate the scanning of the thin resection margins via OCT, rather than including patients with bulky tumours. The limitation of OCT lies in tumour depth. The current generation of this technology, even with the upgrade to four channel systems, still suffers from poor depth of tissue penetration. Our current experience shows that loss of tissue accuracy and definition occurs beyond 2 mm; hence tumours extending deeper than this range are difficult to map in 3 dimensions.

The limited depth of penetration of OCT was not thought to play a role in the results since the poorer results were obtained in the lateral mucosal rather than the deep margins. The significant difference in oral epithelium was found and attributed to the architectural changes and breach of the basement membrane. It is always important to accurately evaluate the extent of the tumour using conventional visual examination and palpation. Computed tomography (CT) and magnetic resonance imaging (MRI) have been used to evaluate tumour extensions. These radiological techniques do not have adequate resolution to accurately measure tumour dimension with certainty. The delineation between the normal tissue and tumour is confusing which makes it difficult to identify most of the tumour within the 5 mm; a problem when managing T₁ and T₂ oral cancers [21].

As this was an immediate ex vivo study, we encountered no problems when assessing tumours involving challengingly accessible anatomical locations. However, we anticipate that there will be some degree of challenge when performing an in vivo study on oral, oropharyngeal or laryngeal tissues. Further modifications of the OCT probe will be required to allow navigation around several hard and soft tissue structures.

OCT examination has the advantage of providing real time, in situ and reproducible measurements. The technique is harmless

to the tissue (non-invasive) and safe to the patient (without ionization radiation). In dermatology, OCT was effective in the diagnosis of benign and malignant skin pathologies [22]. The technique also allows photodocumentation.

When compared to ultrasound, OCT showed good resolution with ability to obtain detailed data concerning the tongue, floor of mouth and larynx [23]. Songra et al. reported on 14 patients with oral carcinoma undergoing ultrasound imaging [24]. They found ultrasonic detection of close surgical margins to have a sensitivity of 83% and a specificity of 63%.

Shintani et al. used ultrasound to assess the thickness of tongue carcinomas preoperatively. Scans and measurements were compared with those from histological sections after resection of the primary tumor. Their study showed significant correlation between the two measures. However, one of the most important inherited problems of the ultrasound in the assessment of the oral cancer is the difficulty to distinguish inflammation from the tumour itself [25]. At this stage and in its current form, OCT cannot differentiate between tumour infiltration and local tissue inflammation. However, OCT is shown to provide a high definition image of tissue within two millimetres of depth, which makes it superior than ultrasound.

Determination of marginal status by frozen section has been reported not only in oral cancer but also in breast and prostate carcinoma [26,27]. This technique is limited by its inability to provide real-time diagnosis; furthermore it has many limitations including low accuracy, limited special stains and studies as well as freezing artifacts.

Shrinkage of the specimen may produce underestimation or overestimation in tumour thickness measurements. Shrinkage usually occurs immediately after tissue resection due to collagen and muscle fibre contraction and during tissue embedding in formalin and preparation before histopathological assessment. Regardless of the type of shrinkage, tissue changes occur volumetrically in three dimensions. Contraction of the vertical collagen and muscle fibres in the connective tissue may increase the actual epithelium thickness. However, contraction of the vertical bundles may not alter epithelium thickness substantially due to difficulty of cell sliding in vertical manner [28]. Johnson et al. [28] reported the mean of loss of depth in tongue specimens following resection to be minimally 30.7%. Our OCT images did not suffer greatly from the shrinkage factor as the scans took place in the immediate ex vivo phase (within 1 h) and the specimens were preserved in saline immediately after they were removed from the patient.

The potential interpretation bias was reduced by the single blind style of the study and the two assessors. This was the most appropriate study design to create a large databank of tumour-free and tumour-involved resection margins. We could not explain the less-than-adequate agreement between the two assessors when it comes to the “lateral margin” scans.

The reason behind some of our false negatives is due to the fact that OCT could only provide accurate tissue definition for 2 mm in depth and certain tumours grow in a pattern than can only be detected at a deeper level (for example iceberg-like shape tumours). Furthermore, in case of a non-cohesive invasive front, even conventional assessments of depth measurements may be inaccurate. In addition, tumors frequently develop finger-like extensions (into the submucosa), which can be hard to identify by OCT. Scanning the central part of the tumour may give an overall idea of the shape and the extension of the tumour.

Two potential weaknesses in our study include the ex vivo nature of the study which tends to affect the OCT image resolution and contrast due to the lack of normal tissue perfusion, and the difficulty to have an exact comparison between the OCT image and the histopathology one due to the shrinkage factor.

6. Conclusion

Optical coherence tomography is an evolving technology. Various clinical studies have suggested that OCT may be of high significance in identifying abnormal tissue. Our study demonstrated that tumour-free and tumour-involved surgical margins could be identified with a good degree of certainty using OCT. The OCT prediction of the tumour-involved margin mainly depends on assessing tumour thickness and architectural changes. Further studies are required to assess tumour margins in vivo.

Conflict of interests

Mr Colin Hopper is an Advisory Board Member at Michelson Diagnostics, Kent, UK. Dr Gordon McKenzie is a Medical Applications Director at Michelson Diagnostics, Kent, UK.

Authors' contributions

All authors designed and performed the study, carried out the literature research and manuscript preparation. All authors were responsible for critical revision of the scientific content and manuscript review. All authors approved the final version of the manuscript.

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